

# Review Report

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## Comparative Summary: Machine Learning & Deep Learning Approaches to Earthquake Magnitude Prediction

*A brief comparative review of three studies (P1, P2, P3) on ML/DL-based seismic magnitude prediction*

### 1. PAPERS REVIEWED

| ID | Title & Authors                                                                                                                          | Venue / Year           | Focus                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------|
| P1 | Comparative Analysis of Decision Tree, Random Forest, SVM, and Neural Network Models for Predicting Earthquake Magnitude — Turino et al. | JUTIF, 2025            | 4-model comparison on global USGS data             |
| P2 | Machine learning for predicting earthquake magnitudes in the Central Himalaya — Tiwari et al.                                            | BIBECHANA, 2025        | RFR/SVR/MLPR comparison for major Himalayan quakes |
| P3 | Prediction of earthquakes in Nepal and the adjoining regions using LSTM — Aryal et al.                                                   | BMC J. Sci. Res., 2024 | PSO-optimized LSTM for Nepal seismic time series   |

### 2. OVERVIEW

The three papers approach earthquake magnitude prediction at different geographic scales and with different model families: **P1** benchmarks four classical/ensemble algorithms — Decision Tree, Random Forest, SVM, and Neural Network — on a large global USGS dataset; **P2** narrows the focus to major (>6 mb) earthquakes in the Central Himalaya, comparing Random Forest, SVR, and MLP regressors; and **P3** applies a Particle-Swarm-optimized LSTM to capture temporal seismic patterns specifically in Nepal. Despite differing regions, dataset sizes, and model architectures, all three arrive at a similar conclusion: ensemble/tree-based methods (Random Forest in particular) and well-tuned sequence models outperform simpler baselines, though every approach still struggles to accurately predict the magnitude of rare, large earthquakes.

### 3. COMPARISON MATRIX

| Criteria       | P1 — 4-Model USGS Study | P2 — Central Himalaya Study            | P3 — Nepal LSTM Study     |
|----------------|-------------------------|----------------------------------------|---------------------------|
| Region / Scope | Global (USGS catalog)   | Central Himalaya (Nepal, India, China) | Nepal & adjoining regions |

|                      |                                                                         |                                                                       |                                                                |
|----------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| Core Approach        | Decision Tree, Random Forest, SVM, and Neural Network compared          | Random Forest, SVR, and MLP Regressor compared                        | LSTM (recurrent network) tuned via Particle Swarm Optimization |
| Prediction Task      | Continuous magnitude regression                                         | Continuous magnitude regression for major (>6 mb) events              | Sequential/time-series magnitude regression                    |
| Dataset Size         | 102,452 events (global USGS)                                            | 2,595 events (1964–2022, ISC catalog)                                 | 2,836 events (2,268 train / 568 test, ISC catalog)             |
| Best Result          | Random Forest: $R^2=0.574$ , MAE=0.217                                  | Random Forest Regressor: closest to actual (e.g., Gorkha 6.24 vs 6.9) | LSTM: MAE=0.279, RMSE=0.373, $R^2=0.429$                       |
| Weakest Model Tested | Neural Network ( $R^2=0.244$ , hyperparameter-sensitive)                | MLP Regressor (near-random, $\sim 4.1$ for all events)                | N/A — single model study                                       |
| Main Limitation      | Relies solely on historical USGS data; no real-time/geological features | Small dataset; all models underestimate large earthquakes             | Underestimates high-magnitude events; limited feature set      |

## 4. PAPER-WISE SUMMARY

### P1 — 4-Model USGS Study (Turino et al., 2025)

Compares Decision Tree, Random Forest, SVM, and Neural Network on 102,452 global USGS earthquake records using features such as latitude, longitude, depth, and station/error metadata. Random Forest achieved the best performance (MAE=0.217, RMSE=0.322,  $R^2=0.574$ ), attributed to its bagging and feature-randomization approach, which reduces overfitting. SVM was a close second ( $R^2=0.531$ ), while Decision Tree and Neural Network lagged behind ( $R^2\approx 0.22$ – $0.24$ ). The main limitation is reliance purely on historical USGS attributes, without incorporating real-time or geological data.

### P2 — Central Himalaya Study (Tiwari et al., 2025)

Focuses specifically on predicting the magnitude of five major 2015 Himalayan earthquakes (including the Gorkha and Kodari events) using Random Forest Regressor (RFR), Support Vector Regressor (SVR), and Multi-Layer Perceptron Regressor (MLPR), trained on 2,595 ISC-cataloged events from 1964–2022. RFR produced predictions closest to actual magnitudes (e.g., 6.24 vs. an actual 6.9 for the Gorkha earthquake) and achieved the highest accuracy (0.16) among the three models. MLPR performed poorly, predicting nearly the same value ( $\sim 4.1$ ) for all five earthquakes regardless of actual magnitude. All models, however, systematically underestimated the larger events.

### **P3 — Nepal LSTM Study (Aryal et al., 2024)**

Applies a Long Short-Term Memory (LSTM) network, with hyperparameters tuned via Particle Swarm Optimization (PSO), to a time-series of 2,836 Nepal-region earthquakes (2,268 for training, 568 for testing) drawn from the ISC catalog. Using engineered temporal and rolling-mean magnitude features, the model achieved MAE=0.279, RMSE=0.373, and  $R^2=0.429$  (cross-validated  $R^2=0.447$ ), successfully capturing general magnitude trends over time. However, like the other two papers, the LSTM consistently underestimates high-magnitude earthquakes (above  $\sim 5.5$ ), reflecting a broader class-imbalance challenge in seismic datasets where large earthquakes are rare.

## **5. KEY TAKEAWAYS**

- Random Forest / ensemble tree-based methods consistently outperform single models (Decision Tree, MLP) and even competitive kernel methods (SVM/SVR) across both global (**P1**) and regional (**P2**) datasets.
- Sequence models like LSTM (**P3**) can capture temporal seismic patterns reasonably well, but their performance depends heavily on hyperparameter tuning (here via PSO) and remains comparable to, not dramatically better than, ensemble regressors.
- A recurring failure mode across all three papers is systematic underestimation of large-magnitude earthquakes — the most consequential events for early warning systems are also the hardest to predict accurately.
- Dataset scale varies dramatically (2,595 to 102,452 events), yet the relative ranking of model families (RF/ensemble > SVM/SVR > NN/MLP) remains fairly consistent.

## **6. CONCLUSION & FUTURE DIRECTIONS**

Together, these studies reinforce that ensemble tree-based methods remain a strong, reliable baseline for earthquake magnitude prediction, whether at global (**P1**) or regional (**P2**) scale, while sequence models like LSTM (**P3**) offer a complementary way to exploit temporal seismic patterns. Future work across all three should focus on enriching feature sets with geological, environmental, and real-time monitoring data, addressing the persistent underestimation of large-magnitude events through imbalance-aware training strategies, and exploring hybrid architectures that combine ensemble learning with deep sequence models.